

Hospital Management System

1. Student Details

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About Me: I am a student in the BS Degree Program at Indian Institute of Technology Madras and also pursuing a dual degree in Computer Science. I am passionate about learning new technologies, especially in the field of web application development, and enjoy exploring tools and frameworks that help build efficient and scalable applications.

2. Project Details

Problem Statement:

Hospitals need efficient systems to manage patients, doctors, appointments, and treatments. Currently, many hospitals use manual registers or disconnected software, which makes it difficult to manage records, avoid scheduling conflicts, and track patient history.

To design a **Hospital Management System (HMS)** web application that allows **Admins, Doctors, and Patients** to interact with the system based on their roles.

Approach:

Developed a web application using **Flask** (backend) and **Vue.js** (frontend). The platform enables users to view doctors by specialization, check their availability, and schedule appointments. It also includes functionality for storing prescriptions and treatment reports for efficient medical record management.

3. AI/LLM Declaration

AI assistance (around 8%) was used primarily for styling and layout refinement, along with inline suggestions and to improve code correctness and robustness.

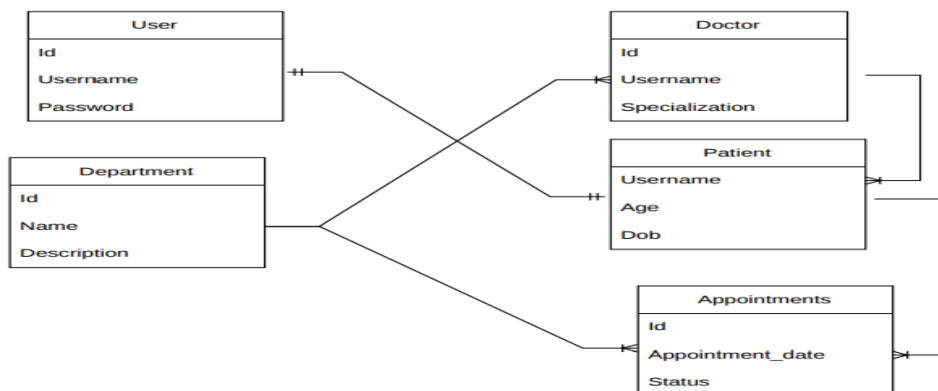
4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Vue3	For reactive UI framework
Celery & Redis	Background task queue

5. Database Schema / ER Diagram

Tables:

1. **User** - stores the user details (Id,username,fullname,address)
2. **Doctor**- stores Doctor profile (Id,username,specialization,experience)
3. **Patient**- stores patient information (Id,username ,age,DOB)
4. **Department**- stores department detail (Name,description)
5. **Appointment**-store appointments (Id,appointment_date,status)



6. API Resource Endpoints

“/api/register”- register a user , POST

“/api/login” - Authenticate user and start session, POST

“/api/patient/histroy” - Fetch medical history for current log user , GET

7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1z5fl9cyq4cV-7QooxQFv-c32thjV4la-/view?usp=sharing>